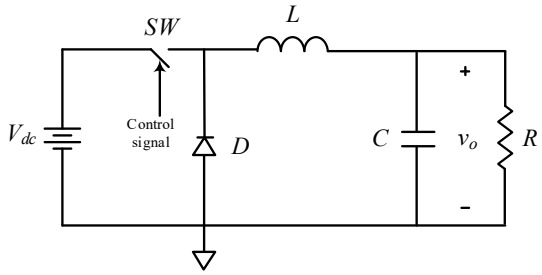
Power Electronics

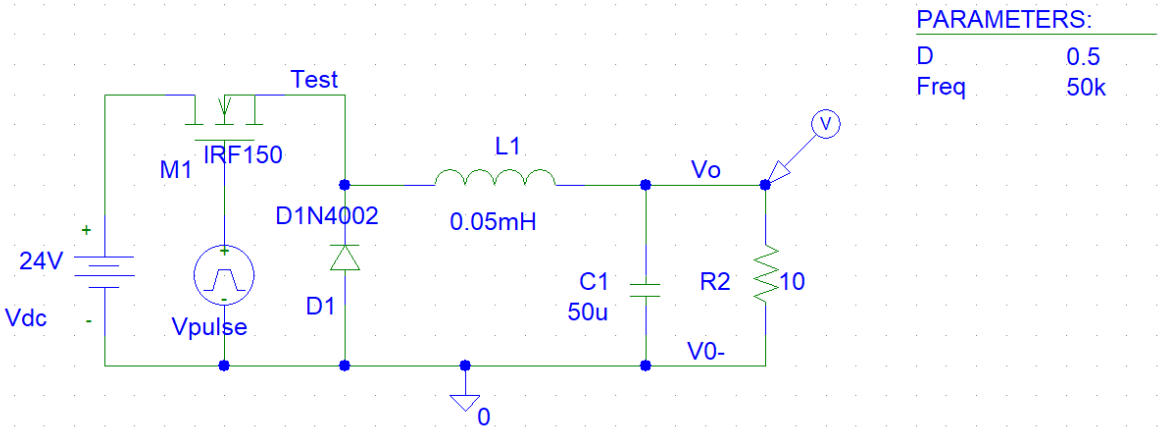
Lab 3 – DC-DC converters

Task1: **Buck converter**

Buil a Buck converter circuit as follows using the following parts

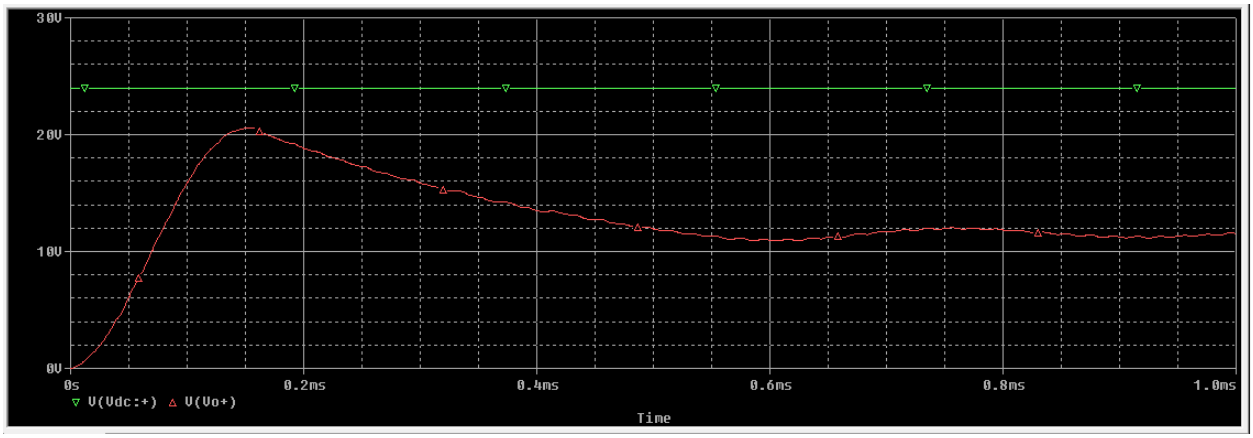
- VDC: input voltage source, 24 V
- VPULSE: to generate control signal for the switch with frequency of f_s = 50 kHz, duty cycle of D = 0.5.
- IRF150 (switch)
- R (load resistance), 10Ω
- L (filter inductance), 0.05mH
- C (filter capacitance), 50µF
- DIN4002 (diode)
- GND_SIGNAL/CAPSYM

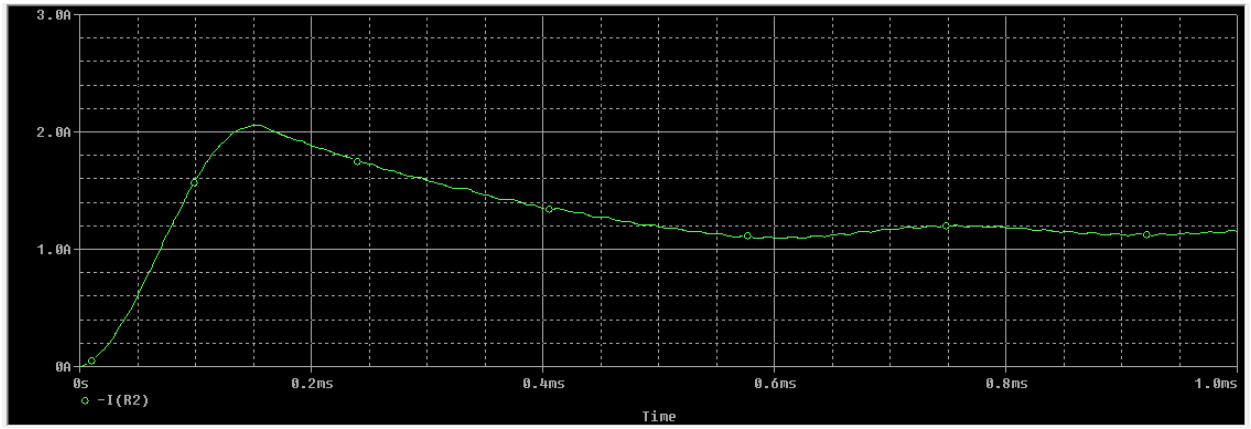




a) Plot the output voltage and output current

(insert plots here)



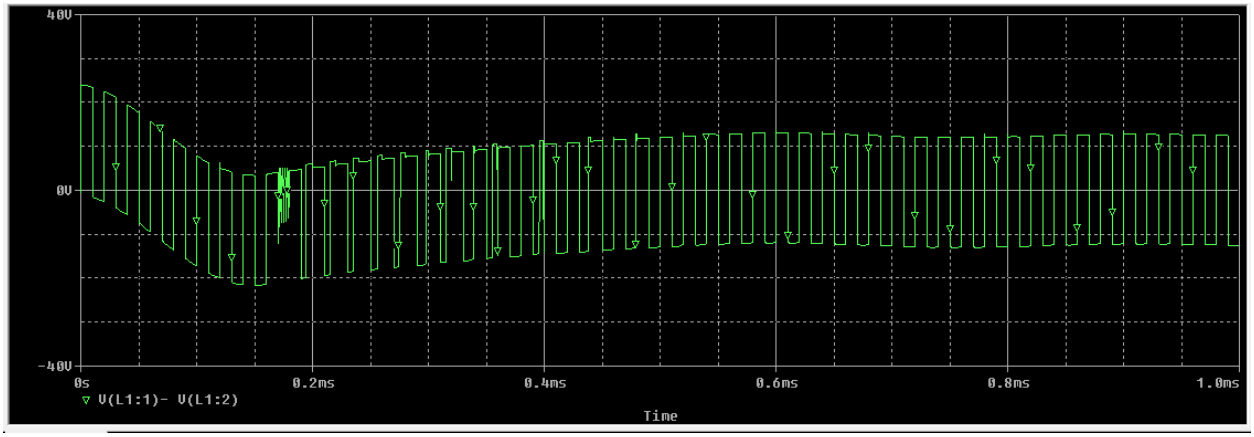


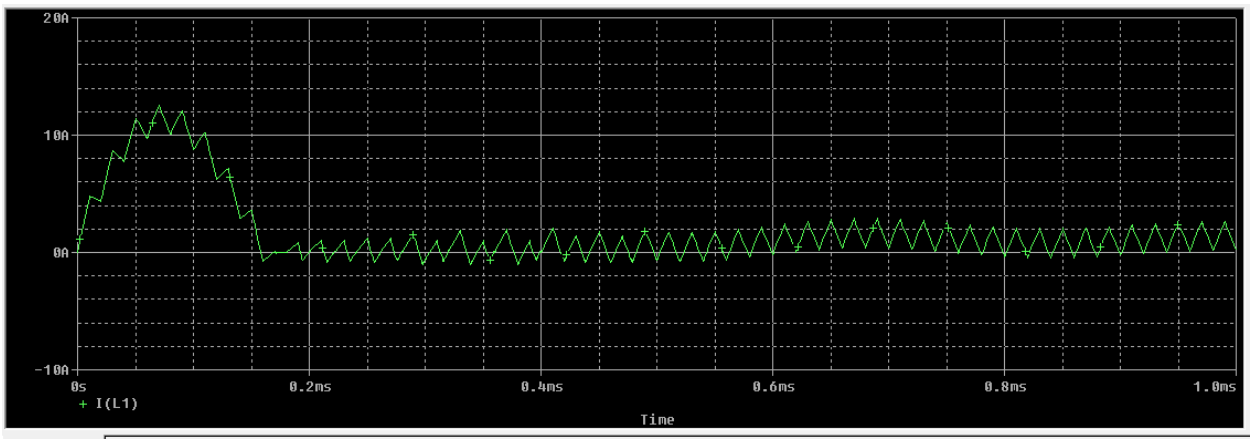
Comment:

The simulation shows the output voltage settling at approximately 12V. This matches the theoretical calculation for a Buck converter. The output current follows the voltage waveform and settles at 1.2A, consistent with Ohm's Law. There is an initial overshoot during the startup transient phase before the system stabilizes.

b) Plot the inductor voltage and inductor current

(insert plots here)

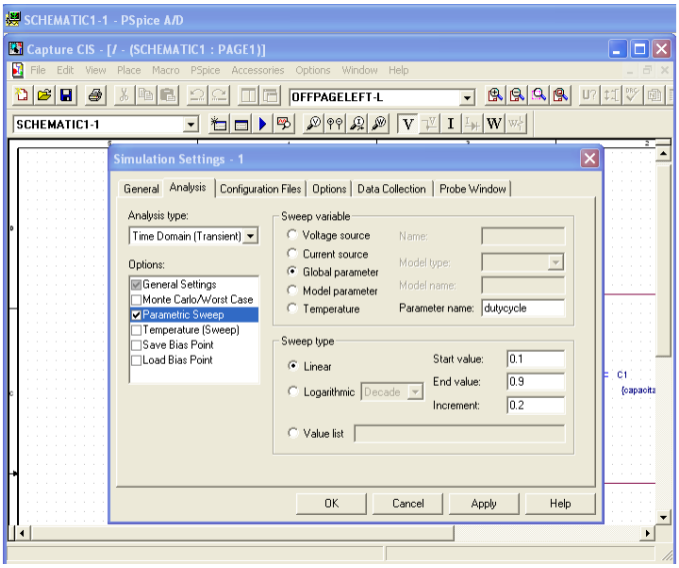




Comment:

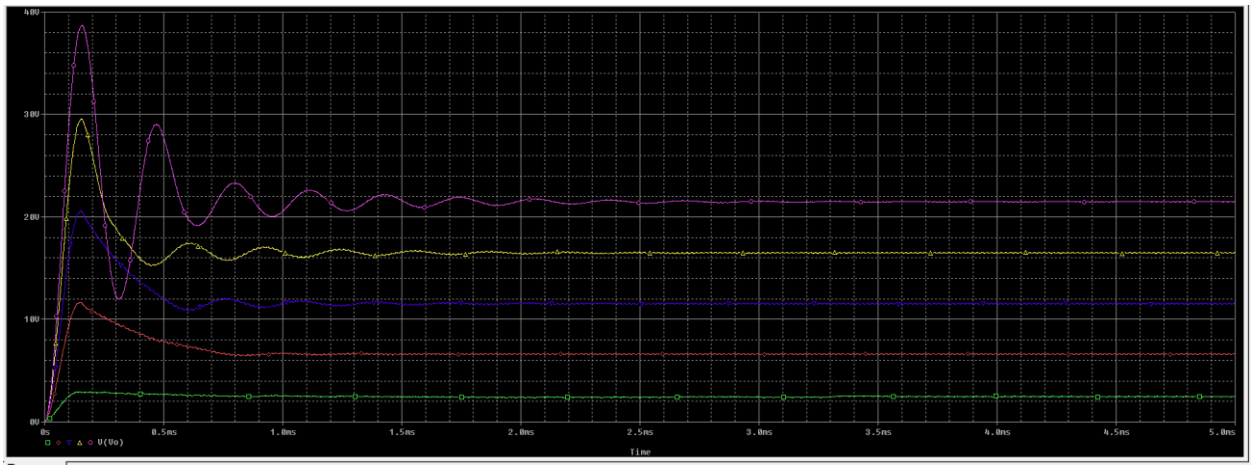
The inductor voltage is a pulse-width modulated (PWM) square wave, alternating between approximately +-12V. The inductor current is a triangular waveform with a DC offset, oscillating around the average load current of 1.2A. The ripple indicates the continuous charging and discharging of the inductor during switching cycles.

c) For getting plots for load voltage at different values of duty cycle use “Parametric sweep” from “Simulation Parameters Setttings” and the put “Duty cycle” as the parameter name. Specify its start value, end value, and increment.



Plot the output voltage for different values of above duty cycle

(insert plots here)

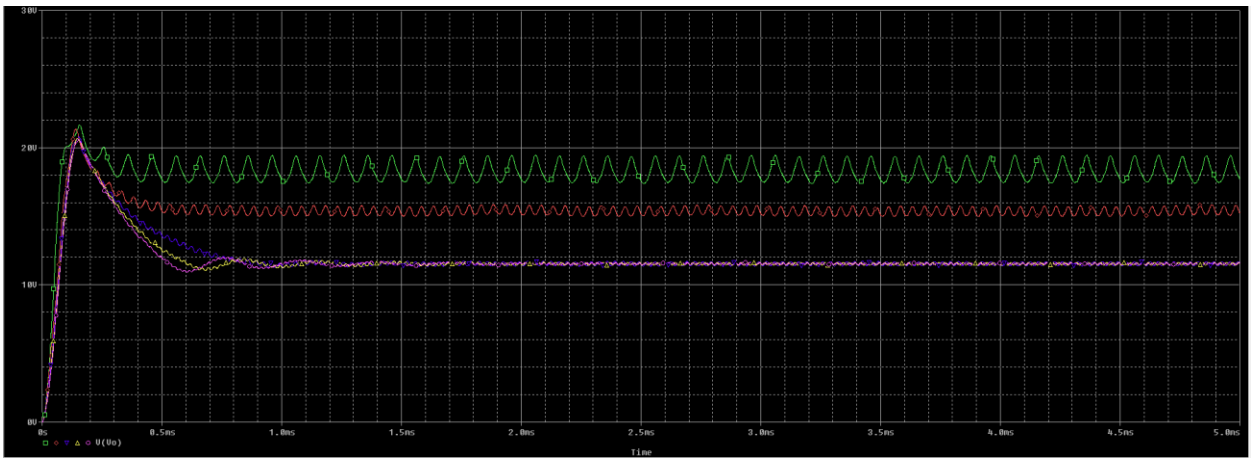


Comment:

As the duty cycle D increases from 0.1 to 0.9, the steady-state output voltage increases proportionally. The highest curve corresponds to D=0.9, while the lowest curvecorresponds to D=0.1

d) Similar, vary the switching frequency from 10kHz to 50kHz with increment of 10kHz, plot the output voltages

(insert plots here)



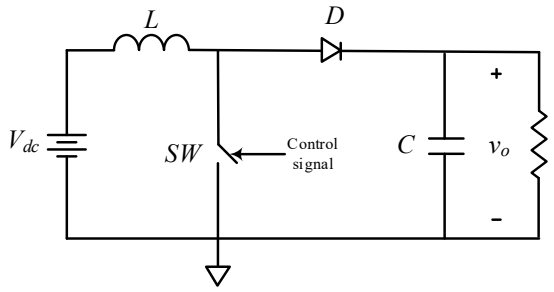
Comment:

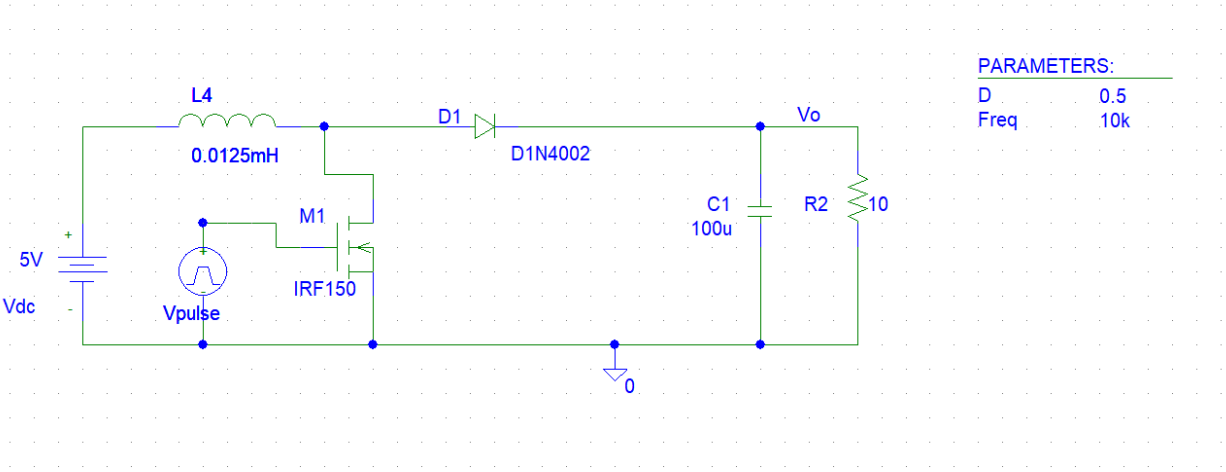
The plot shows that changing the frequency affects the output ripple and transient response. Lower switching frequencies result in significantly higher voltage ripple because the capacitor must sustain the load for a longer duration during the "off" state. Higher frequencies produce a smoother DC output with less ripple.

Task 2. Boost converter

Buil a Buck converter circuit as follows using the following parts

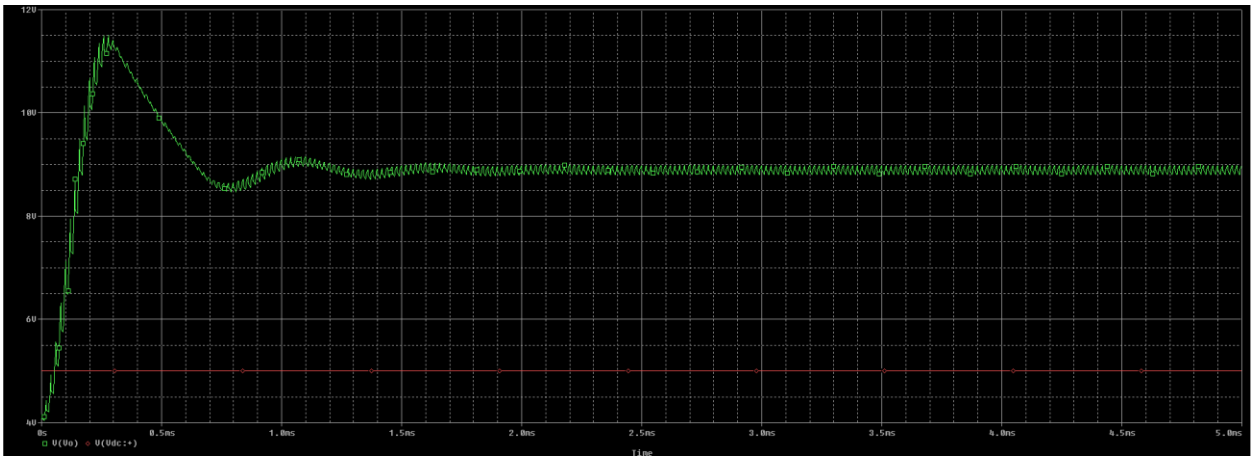
- VDC: input voltage source, 5 V
- VPULSE: to generate control signal for the switch with frequency of f_s = 50 kHz, duty cycle of D = 0.5.
- IRF150 (switch)
- R (load resistance), 10Ω
- L (filter inductance), 0.0125mH
- C (filter capacitance), 100μF
- DIN4002 (diode)
- GND_SIGNAL/CAPSYM





a) Plot the output voltage and output current

(insert plots here)



Comment:

With an input of 5V and a duty cycle of 0.5, the output voltage settles around 1V. This confirms the Boost converter equation. The output current settles at 1A, matching the expected load current.

b) Plot the inductor voltage and inductor current

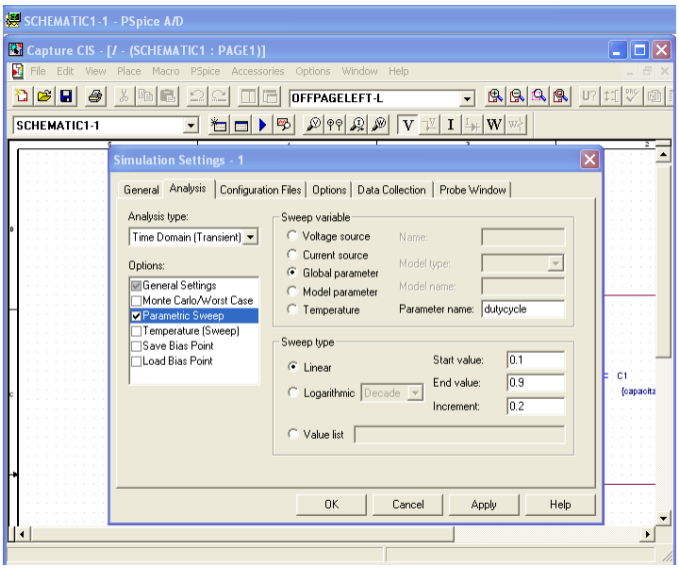
(insert plots here)



Comment:

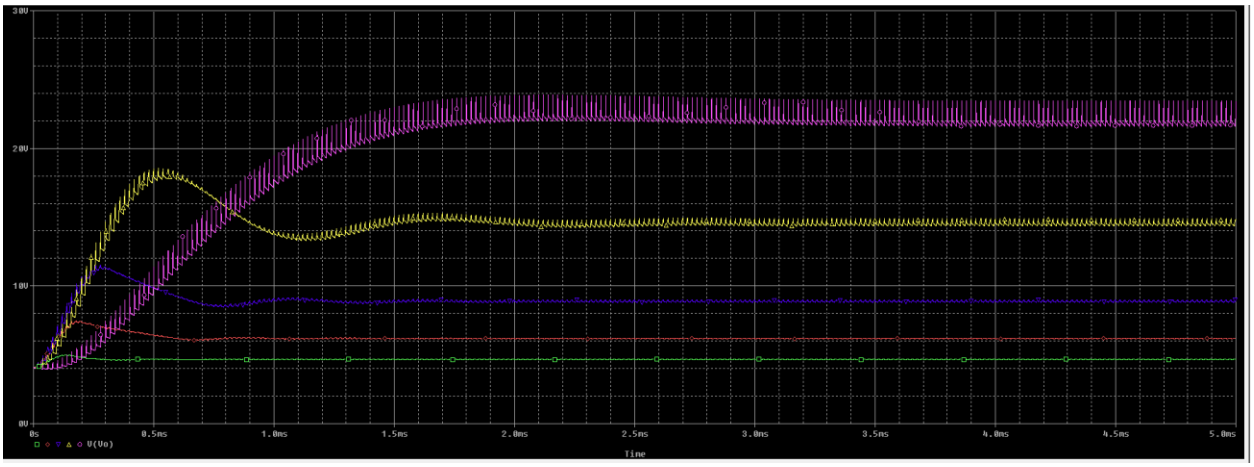
The inductor current waveform is triangular and centers around **2A**. This is expected because, in a Boost converter, the input current is higher than the output current

c) For getting plots for load voltage at different values of duty cycle use “Parametric sweep” from “Simulation Parameters Setttings” and the put “Duty cycle” as the parameter name. Specify its start value, end value, and increment.



Plot the output voltage for different values of above duty cycle

(insert plots here)

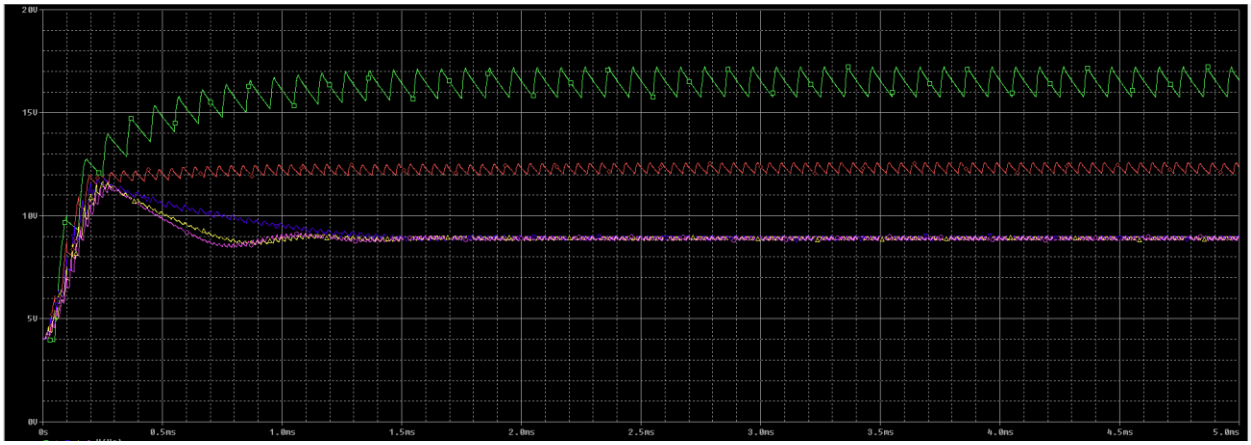


Comment:

The sweep shows that increasing the duty cycle results in a non-linear, rapid increase in output voltage. At low duty cycles, the output is close to the input (5V). As D approaches 0.9the voltage boosts significantly high, demonstrating the high sensitivity of Boost converters at high duty cycles.

d) Similar, vary the switching frequency from 10kHz to 50kHz with increment of 10kHz, plot the output voltages

(insert plots here)



Comment:

The plot demonstrates that lower switching frequencies cause much larger current ripple in the inductor. The higher frequency traces show much tighter, smaller ripple.